



FUSE ADVISORS

A New Approach

Developing Mining Projects Using a Multiple-Criteria Decision Making Structure

CIM 2023 Montreal

May 4th, 2023



FUSE ADVISORS

OUR TEAM

Partner



Ryan Weymark
President

Mr. Weymark is an energetic Professional Engineer (P. Eng.) with experience in mining, heavy civil and infrastructure projects.

Ryan has experience working on projects as a consultant, contractor and owner's representative at various stages, including technical due-diligence, economic studies (pre-feasibility to feasibility), permitting (provincial and federal) and construction.

Consultant



Ben Collins
Senior Technical Consultant

Mr. Weymark is an energetic Professional Engineer (P. Eng.) with experience in mining, heavy civil and infrastructure projects.

Ryan has experience working on projects as a consultant, contractor and owner's representative at various stages, including technical due-diligence, economic studies (pre-feasibility to feasibility), permitting (provincial and federal) and construction.



FUSE ADVISORS

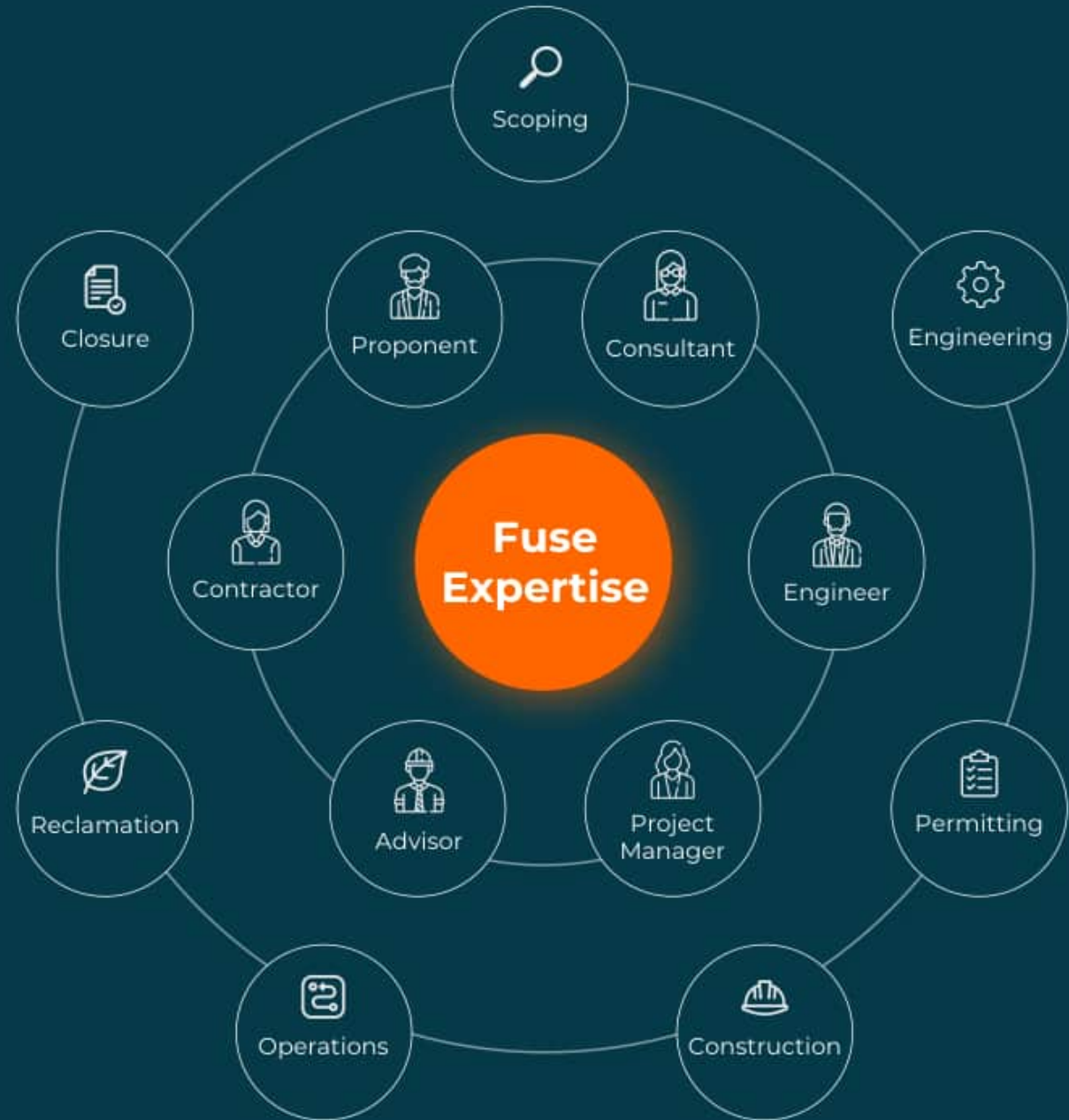
OUR STRENGTHS

Fuse takes a curated **client-first approach** to delivering high-quality services, tailored to the distinctive needs of our clients.

Our experience allows for the effective management of complex projects through the removal of the functional silos between engineering, permitting, construction, and operations.

Optimized outcomes are achieved by early identification of project risks and the strategic implementation of timely mitigations.

With our highly diversified experience in project management and advisory, we provide **integrated Owner's team support** without bias or competing interests.





FUSE ADVISORS

WHO ARE WE?

Fuse Advisors are passionate experts able to provide an integrated approach to **project management** and **advisory**



**Project
Manager**



**Owner
Representatives**



**Technical
Advisors**



**Project Controls
Specialists**



**Engineering
Managers**



**Market
Researchers**





FUSE ADVISORS

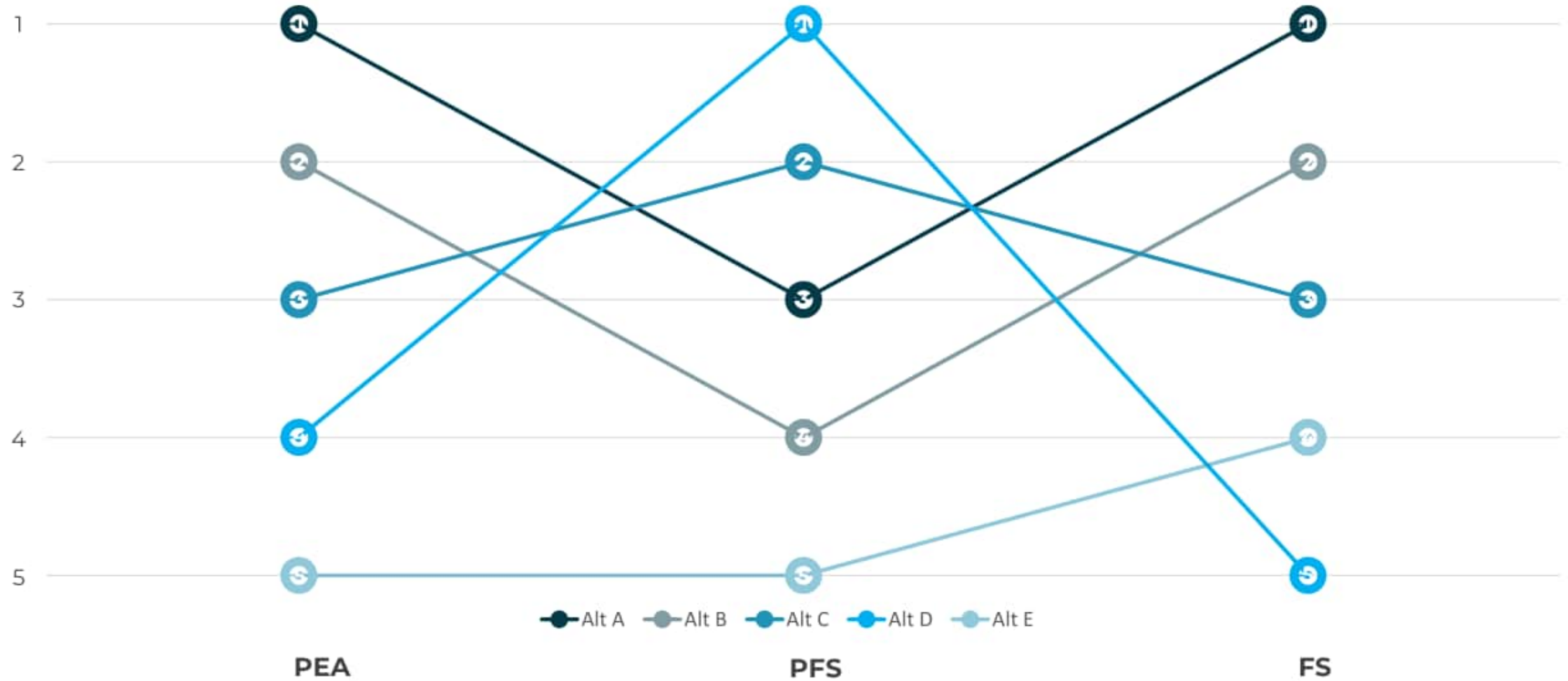
PROJECT DEVELOPMENT PROCESS





FUSE ADVISORS

CHANGING RANKINGS





FUSE ADVISORS

THE PROBLEM



Mining developers must make design assumptions early on with very little information.

1



Often these assumptions get carried forward without realizing how little information backs these decisions

2



FUSE ADVISORS

THE PROBLEM



Mining developers must make design assumptions early on with very little information.

1



Often these assumptions get carried forward without realizing how little information backs these decisions

2



FUSE ADVISORS

THE SOLUTION



Create multiple-criteria decision making approaches and tracking tools as early as possible to track technical data, Indigenous collaboration, and key criteria, to make faster decisions.

3



FUSE ADVISORS

THE SOLUTION



Create multiple-criteria decision making approaches and tracking tools as early as possible to track technical data, Indigenous collaboration, and key criteria, to make faster decisions.

3





FUSE ADVISORS

SESSION GOALS



1 Learn and Appreciate Multiple Criteria Decision Making



2 How it's being applied in the Mining Industry



3 Where Fuse Advisors is seeing its potential for the Mining Industry





FUSE ADVISORS

SEEING THE MINING INDUSTRY POTENTIAL

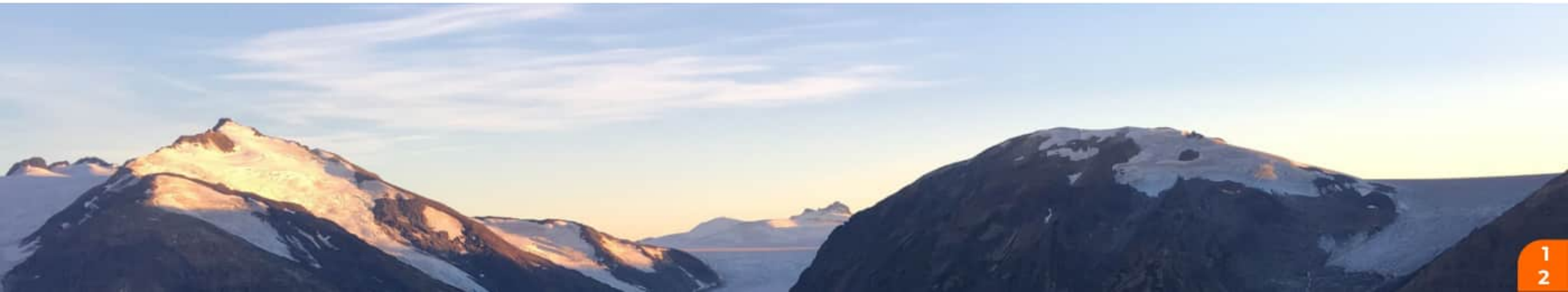
Implementing wants, needs, and values of Indigenous communities **(MCDM)**



Tracking uncertainty in decision making **(MAA)**



Providing project managers with tools to implement multiple criteria





FUSE ADVISORS

MCDM, MAAs, MCDAs

What's The Difference?

-  Multiple Criteria Decision Making (MCDM) and Multiple-Criteria Decision Analysis (MCDA) are the same
-  Multiple Accounts Assessments (MAAs) is a type of an MCDM/MCDA seen during alternative assessments for key technologies
-  They are structured approaches to select alternatives based on key criteria and the decision maker values





FUSE ADVISORS

MULTIPLE CRITERIA PROCESS FLOW

Investigation

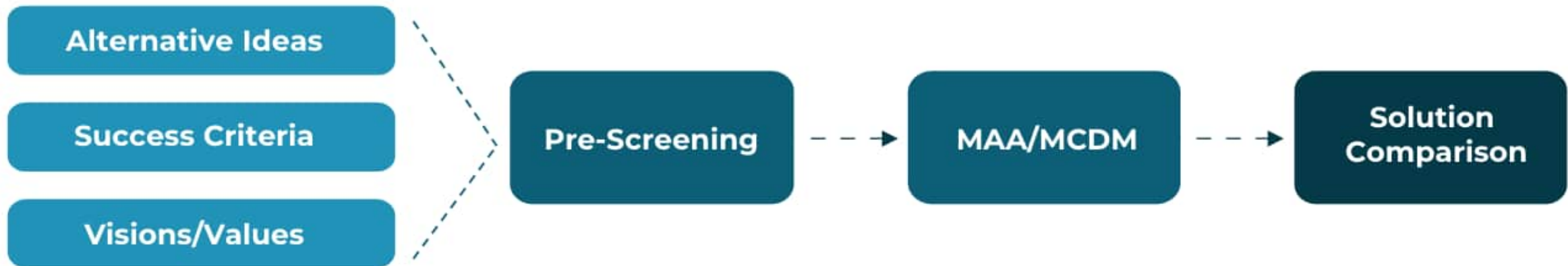
- Investigate the potential project alternatives, criteria, and values for the decisions
- Can be through workshops, surveys, and strategic discussions.

Analyze

- Utilize MCDM techniques to create relative weights of criteria
- Analyze how each alternative idea compares with our criteria for success

Decision

- Score and compare the potential solutions
- Discuss and analyze the results
- Iterate on previous steps if needed
- Decide on best solution





FUSE ADVISORS

PRE-SCREENING VS. MCDM

Project Alternative	Level of Assessment
Primary mining method	Screening
Underground mining method	Screening
Process Plant Location	Screening
Process Stripping Method	Screening
Final product type	Screening
Mine waste storage	MAA
Power supply type	Screening
Fuel delivery method	Screening
Camp Location	Screening
Airstrip Location	Screening
On-site road alignment	Screening
Effluent treatment technology	MAA
Fresh water supply – source/location	Screening
Waste disposal (domestic, industrial, hazardous)	MAA

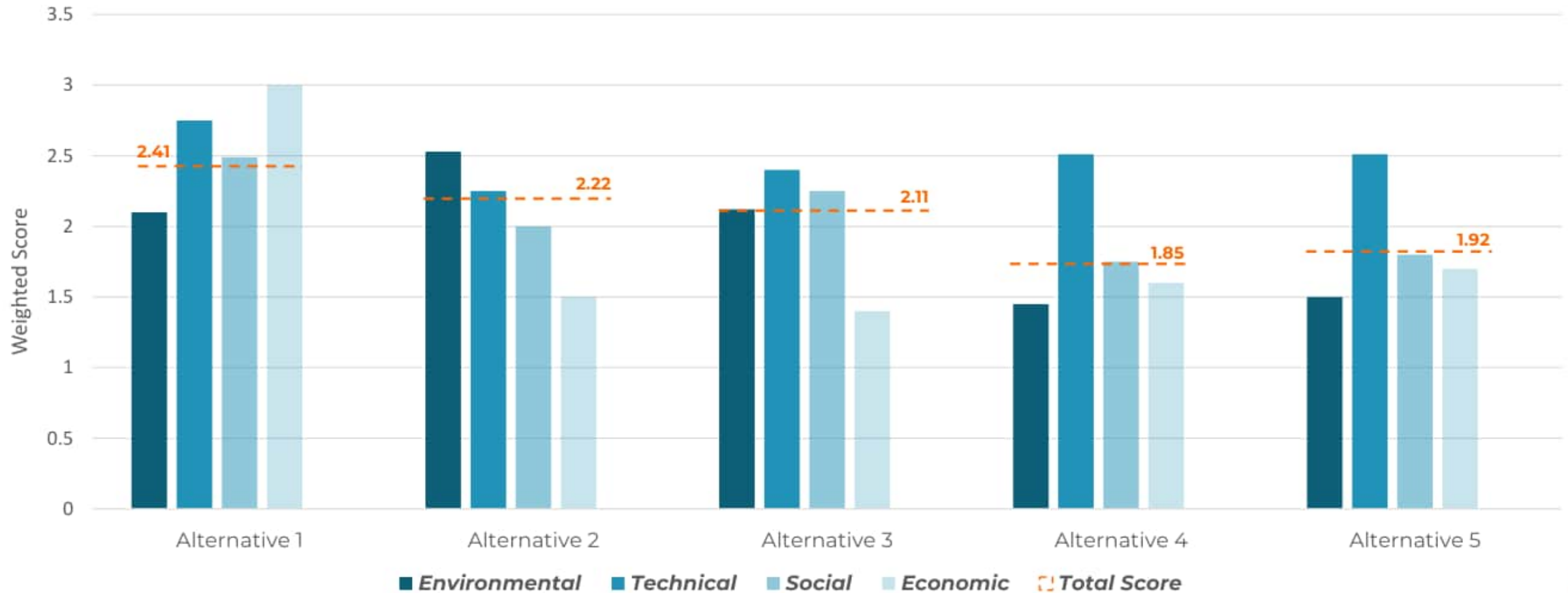




FUSE ADVISORS

MAA OUTPUT

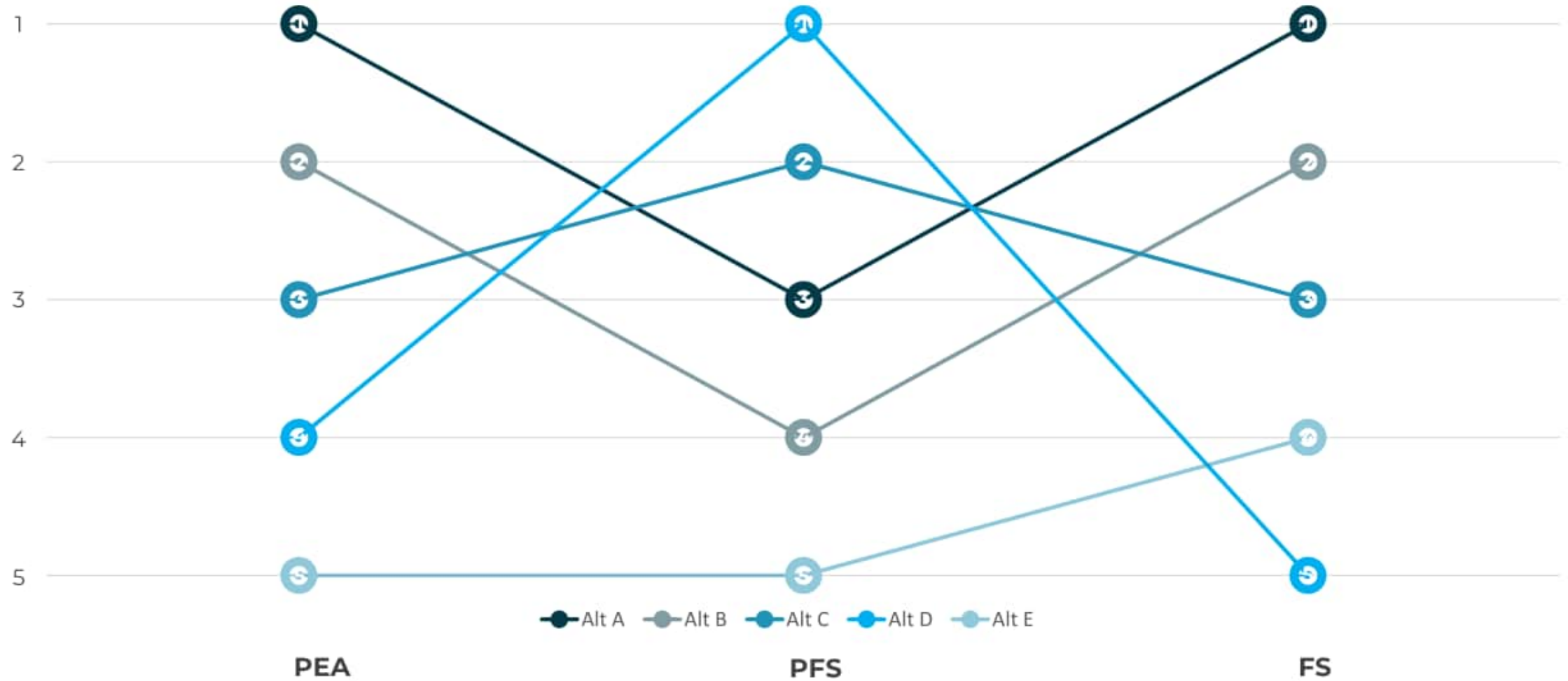
For Mine Waste Management Approaches





FUSE ADVISORS

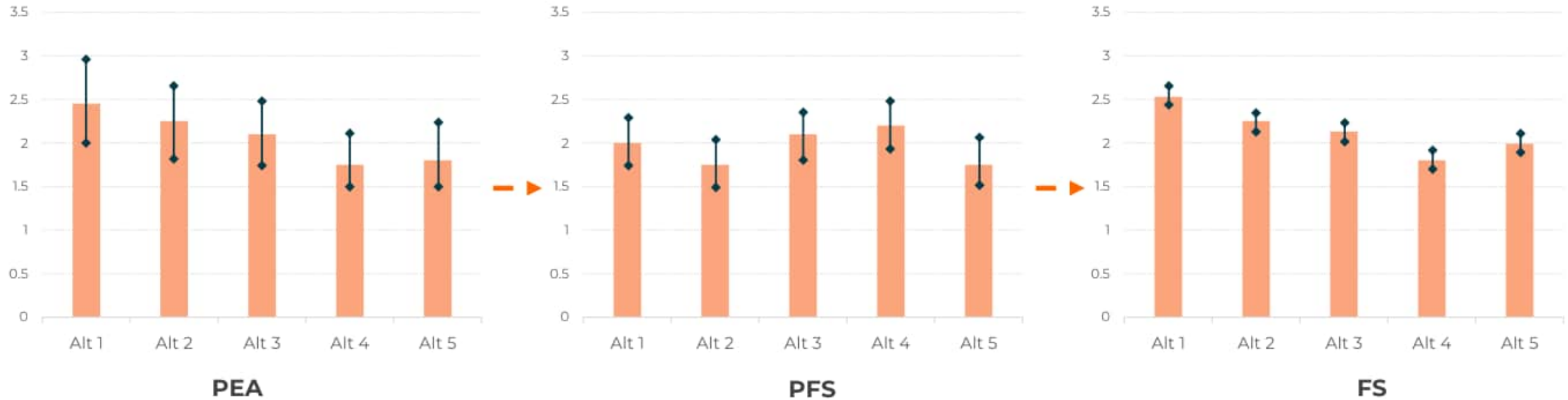
CHANGING RANKINGS





FUSE ADVISORS

HOW AN MCDM CAN CHANGE OVER TIME





FUSE ADVISORS

TECHNICAL APPLICATIONS

Power Type, Location, And Distribution



Key Criteria

- GHGs
- Project Economics
- Local Environmental Impacts
- Indigenous Community Consultation



Alternative Considerations

- Grid
- Diesel
- LNG.





FUSE ADVISORS

COLLABORATING WITH INDIGENOUS COMMUNITIES



At its most basic level, an MCDM tool can bring together different value systems for a decision-making process.



The development of values, key criteria, and alternatives is an opportunity to bring the local community's voice into project decision making.



MCDMs can provide a framework to collect, track, and incorporate key project issues and risks for Indigenous communities.





FUSE ADVISORS

FINAL THOUGHTS

Decision Making Is A Subjective Process.

MCDM approaches provide a structured approach to this subjective process.



It helps communicate how values, criteria, and alternatives interrelate for project decision making.



Too often, MCDMs occur too late in development, when it is too expensive to change from one alternative to another. They need to be applied as early as possible.





FUSE ADVISORS

Thank You

CIM 2023 Montreal
May 4th, 2023